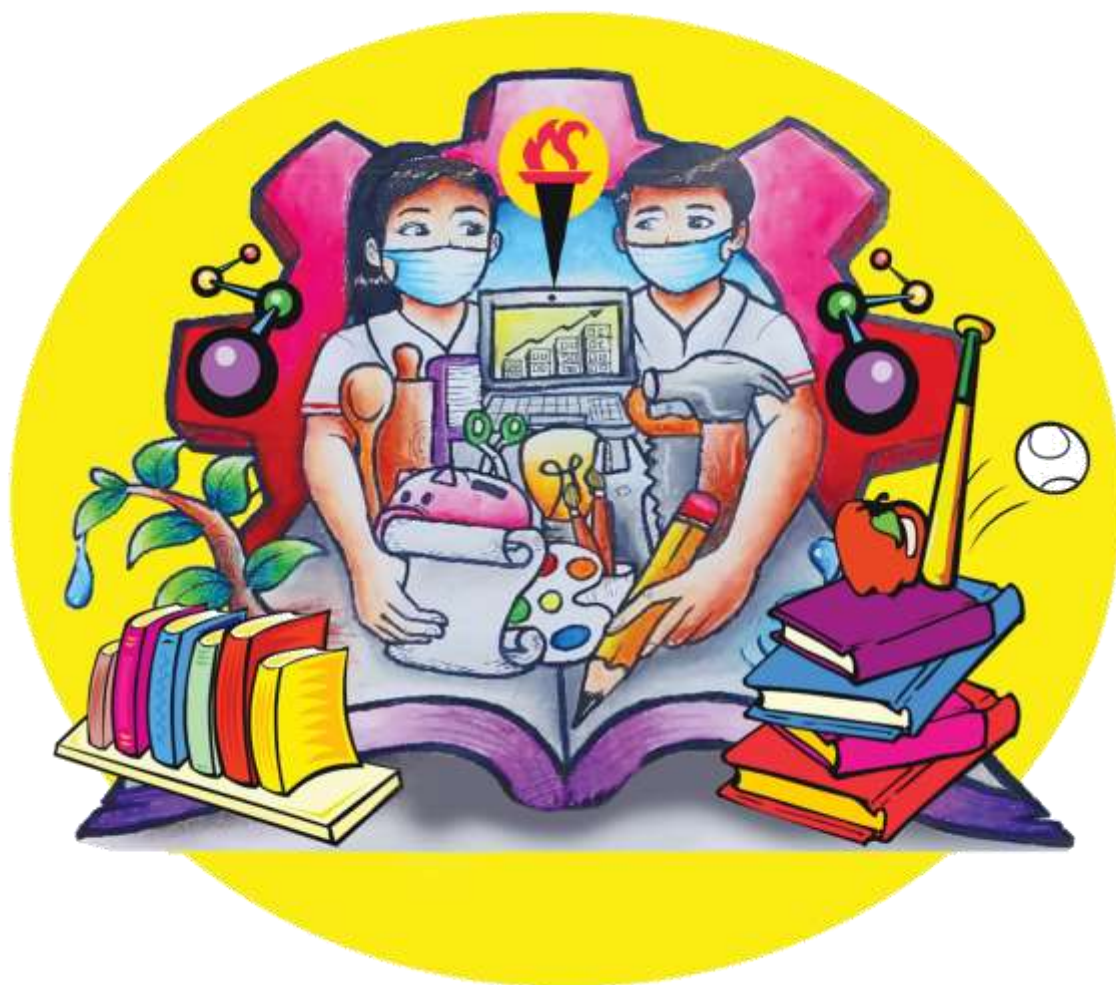


Senior High School

General Physics II

Second Semester (Final) – Module 1

Induced EMF and Induced Current



S L M

SELF-LEARNING MODULE

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Introductory Message

For the facilitator:

Welcome to the General Physics II 12 Self-Learning Module on Induced EMF and Induced Current!

This module was collaboratively designed, developed and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners into guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st century skills while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:



Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator, you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the General Physics II 12 Self-Learning Module on Induced EMF and Induced Current!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

CONTENT STANDARD

The learner demonstrates understanding of magnetic induction and Faraday's Law.

PERFORMANCE STANDARD

The learners should be able to use theoretical and, when feasible, experimental approaches to solve multi-concept, rich-context problems involving magnetic and electric field.

LEARNING COMPETENCY

1. Identify the factors that affect the magnitude of the induced emf and the magnitude and direction of the induced current (Faraday's Law). **(STEM_GP12EM-IVa-1)**
2. Compare and contrast electrostatic electric field and non-electrostatic/induced electric field. **(STEM_GP12EM-IVa-3)**
3. Calculate the induced emf in a closed loop due to a time-varying magnetic flux using Faraday's Law. **(STEM_GP12EM-IVa-4)**

LEARNING OBJECTIVES:

At the end of the module, you should be able to:

1. Determine the factors affecting the induced emf and induced current.
2. Distinguish electrostatic electric field and induced electric field.
3. Calculate the induced emf in a closed loop using Faraday's Law.

INTRODUCTION

What will you do if there is brownout and you are catching up for the deadlines of the submission of your research paper and other school-related requirements? I'm sure you really need sufficient light to efficiently accomplish your tasks. This module will give you an idea on how to produce electricity even during brownout.

You will learn from this module about electromagnetic induction where induced voltage and induced current are produced due to the changing magnetic field.

PRE-TEST

Directions: Read each item carefully and identify what is asked or described. Encircle the letter of the correct answer.

1. What does emf stand for?
 - A. Electromotive force
 - B. Electro mobile force
 - C. electromagnetic factor
 - D. electromagnetic force
2. Which of the following is NOT true about induced emf?
 - A. It is caused by a constant magnetic field.
 - B. It is directly proportional to induced current.
 - C. It is always opposite to the direction of magnetic flux.
 - D. It is only produced by the motion perpendicular to the field.
3. A wire loop is being pulled through a uniform magnetic field. Which of the following statement is TRUE?
 - A. The induced current is constant.
 - B. There is no induced current produced.
 - C. The direction of induced current is clockwise.
 - D. The direction of induced current is counter clockwise.
4. Doubling the number of loops of wire in a coil produces what kind of change on the induced emf, assuming all other factors remain constant?
 - A. The induced emf is twice as much.
 - B. The induced emf is 3 times as much.
 - C. The induced emf is 4 times as much.
 - D. The induced emf remains the same.
5. In the demonstration where a magnetic north pole is brought near a loop, which is TRUE about the induced current if we use the same speed but switch to the south end of the magnet?
 - A. The magnitude and direction are the same.
 - B. The magnitude is the same but the direction is reversed.
 - C. The magnitude is now smaller and in the opposite direction.
 - D. The magnitude is now larger and in the same direction as before.

6. Faraday's law of induction states that
 - A. the direction of the induced emf is such that it opposes the change of flux.
 - B. the magnitude of the induced emf is directly proportional to the rate of change of flux.
 - C. the magnitude of the induced emf is inversely proportional to the rate of change of flux.
 - D. the magnitude of the induced emf is directly proportional to the square of the rate of change of flux.

7. A loop lies in a magnetic field, with the field perpendicular to the plane of the loop. Which does NOT induce a current in the loop?
 - A. Increasing the size of the loop.
 - B. Changing the direction of the field.
 - C. Rotating the loop around the magnetic field.
 - D. Sliding the loop perpendicular to the field vectors.

8. When the north pole of a magnet is moved into a coil, the direction of the induced current in the coil as seen from the magnet side is

A. Clockwise	C. Clockwise then counter clockwise
B. Counter clockwise	D. Counter clockwise then clockwise

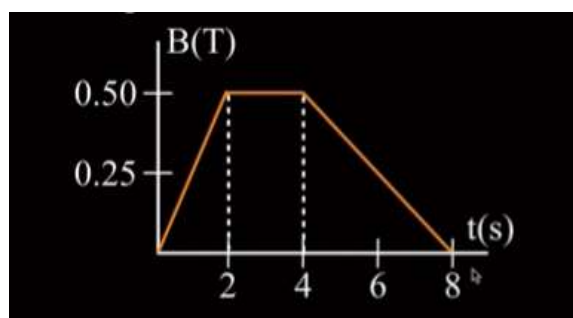
9. The following are factors involved in the induced emf of a coil of wire EXCEPT
 - A. The induced emf is directly proportional to N , the total number of turns in the coil.
 - B. The induced emf is directly proportional to A , the area of cross-section of the coil.
 - C. The induced emf is directly proportional to Δt , the time of changing magnetic flux.
 - D. The induced emf is directly proportional to B , the strength of the magnetic field in which the coil is rotating.

10. Which of the following is NOT true about induced electric field?
 - A. It is caused by the changing magnetic field.
 - B. Its work or energy difference depends on path.
 - C. Its field $\oint \mathbf{E} \cdot d\mathbf{l}$ is not equal to zero, therefore electric potential has no meaning.
 - D. Its force is conservative because it does net work in moving a charge over a closed path.

11. The following are the similarities of electrostatic electric field and induced electric field EXCEPT
 - A. their field lines form closed loops.
 - B. their field lines never cross each other.
 - C. their field exerts a force, $F = qE$ on charged particle.
 - D. their field is strong when the field lines are close together.

12. The coil of wire containing **two loops** with a radius of **6 cm**. is in a **1.8 T** magnetic field. The plane of the coil of wire forms **60°** with the magnetic field. The loop is removed from the magnetic field over a time of **0.25 s**. What is the induced emf in the loop? (Hint: The area of the circular coil is **$A = \pi r^2$**)
- A. $4 \times 10^{-2} \text{ V}$ C. $7 \times 10^{-2} \text{ V}$
 B. $6 \times 10^{-2} \text{ V}$ D. $8 \times 10^{-2} \text{ V}$

For questions #13-15: The magnetic field through a single loop of wire, **10 cm** in radius and with a resistance of 7.5 ohms changes with time as shown in the graph below.



13. What is the induced emf in the loop of wire from $t = 0$ second to $t = 2$ seconds?
- A. $7 \times 10^{-3} \text{ V}$ C. $7 \times 10^{-2} \text{ V}$
 B. $8 \times 10^{-3} \text{ V}$ D. $8 \times 10^{-2} \text{ V}$
14. What is the induced emf in the loop of wire from $t = 2$ seconds to $t = 4$ seconds?
- A. 0 V B. 0.25 V C. 0.5 V D. 2 V
15. What is the induced emf in the loop of wire from $t = 4$ seconds to $t = 6$ seconds?
- A. 0 V B. $2 \times 10^{-3} \text{ V}$ C. $4 \times 10^{-3} \text{ V}$ D. $5 \times 10^{-3} \text{ V}$

“Congratulations you made it!

If you are confident that all your answers are correct, very good! This shows that you already know much about the topics in this module. You may still study the module to review what you already know. Who knows, you might learn a few more new things as well.

If you think you got a low score, don't feel bad. This means that this module is for you. It will help you to understand some important concepts that you can apply in your daily life. If you study this module carefully, you will learn the answers to all the items in the test and a lot more!

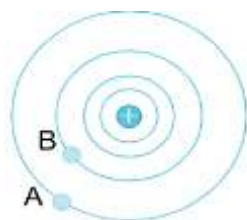
Are you ready? Let's go!”

REVIEW OF THE PREVIOUS MODULE

In Module 2.2, you learned how to solve problems involving electric charges, forces, and fields in charged pendulums, electrical breakdown of air, and electric inkjet printers. Let's have a recap of what you have learned by answering the questions below.

Direction: Answer the following questions.

1. If 22 J of work is done by an external force to move a charge from an electric potential of 9V to a potential of 2 V,
A. what is the change in electric potential energy?
B. what is the electric potential difference across which the charge is moved?
2. Test charge A and B are moved relative to a positive source charge as shown in the figure below. Assume that both test charge A and B are positive and they have the same magnitude.



- A. Which charge has greater electric potential?
- B. Which charge has greater electric potential energy?
- C. Which charge has given more work?

PRESENTATION OF THE NEW MODULE

This module primarily deals with Faradays law of induction. In this module you will learn the factors affecting the induced electromotive force (emf) and induced current; the distinction between electrostatic electric field and induced electric field; and calculation of the magnitude of electromotive force due to the rate of change of magnetic flux.

ACTIVITY: Determining the factors affecting induced emf

- Physicists have conducted experiments to determine the possible factors affecting the induced emf. Below are the data they have gathered from their experiments.

A. Induced emf vs. Number of turns of coil

No. of turns (N)	5	7	9	11	13	15
Induced emf (mV)	0.04	0.05	0.06	0.07	0.08	0.09

Direction: Based on the data presented above, make a line graph of the induced emf against the number of turns.

B. Induced emf vs. Cross-sectional area of coil

- Two square solenoids of different cross-sectional areas were connected in series to the signal generator. The solenoids both had the same number of turns per unit length but one had a larger cross-sectional area than the other.
- Ten turns of the copper wire were wired tightly around the middle of each solenoid. One copper wire was connected to channel 1 of a dual trace CRO and the other to channel 2. The solenoids were kept well apart from each other.
- The signal generator was turned on and set to 1 kHz. The CRO setting was adjusted to display both whole traces on its screen.
- The time base of the CRO was switched off. The length of the vertical trace shown on the CRO was recorded, which represented the induced peak-to-peak emf in the copper coil for each solenoid.

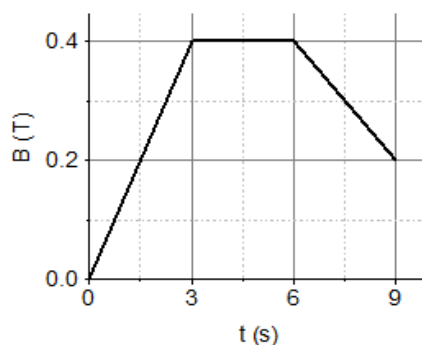
Solenoid	Large	Small
Induced emf (mV)	130	80

Direction: Based on the data presented in the table above, compare the magnitude of induced emf in a solenoid with large cross-sectional area and small cross-sectional area.

- What happens to the magnitude of the induced current when cross-sectional area of the solenoid is greater?
- What happens to the magnitude of the induced current if the cross-sectional area of the solenoid is smaller?

C. Induced emf vs. Rate of change of magnetic flux

In a certain experiment, a magnetic field passes through a single stationary wire loop with an area of $7.85 \times 10^{-3} \text{ m}^2$ and its magnitude changes in time according to the graph below. The direction of the field remains constant however. There are three equal time intervals indicated in the graph: **0 - 3.0 s**, **3.0 - 6.0 s** and **6.0 - 9.0 s**. Note that the number of loop, **N = 1**. Complete the table below by filling in the correct answers based on the graph.



Study.com/academy/answer

Time interval (in unit s)	Change in magnetic field $(\Delta B = B_{\text{final}} - B_{\text{initial}})$ (in unit Tesla, T)	Rate of change in magnetic flux $\Delta \phi = A \Delta B$ (in unit weber, Wb)	Induced Emf $\text{emf} = \frac{N \Delta \phi}{\Delta t}$ (in unit volt, V)
0 s – 3.0 s			
3.0 s – 6.0 s			
6.0 s – 9.0 s			

ANALYSIS

Direction: Answer the following questions on your paper.

A. Induced emf vs. Number of turns of coil

1. What is the graph of induced emf versus number of turns of coil?
2. What does the graph tell you about?
3. What happens to the induced emf when the number of turns of coil increases?
4. What happens to the induced emf when the number of turns of coil decreases?
5. What is the relationship between induced emf and the number of turns of coil?

B. Induced emf vs. Cross-sectional area of the solenoid

1. What happens to the magnitude of the induced current when cross-sectional area of the solenoid is greater?
2. What happens to the magnitude of the induced current if the cross-sectional area of the solenoid is smaller?
3. What is the relationship between the induced emf and the cross-sectional area of the solenoid?

C. Induced emf vs. Rate of change of magnetic flux

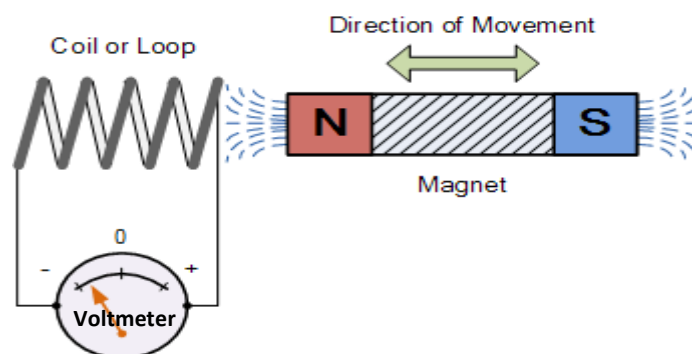
1. What happens to the magnetic flux if the change in magnetic field increases?
2. What happens to the induced emf if the change of magnetic flux increases? Decreases?
3. What is the change of magnetic flux if magnetic field does not change (magnetic field is constant)?
4. What is the magnitude of induced emf if there is no change in magnetic flux?
5. What is the relationship between induced emf and the rate of change in magnetic flux?
6. How is time in changing the magnetic flux related to the magnitude of the induced emf?

Good job! I told you, you can do better! Now, it is time for you to know it all. Read on.

ABSTRACTION

Magnet has its own magnetic field containing numerous field lines. If this field is disturbed by any means, then it would change consequently because its field lines are cut. The more cut field lines are, the greater is the change in magnetic field. The field can be changed by moving a magnet toward or away from the coil, moving the coil into or out of the magnetic field and rotating the coil relative to the magnet. Furthermore, when the strength of the magnetic field changes, there is also a change in magnetic flux. The change in magnetic flux due to a change in magnetic field causes an induced electromotive force (emf) or what we normally called voltage.

A changing magnetic flux creates an emf in a circuit as shown by the figure below.



Electromagnetic Induction and Faradays Law (electronics-tutorials.ws)

Figure 1: A magnet is moved into and away from the wire loop. The emf is induced because the magnetic field lines of the magnet are cut.

Factors affecting the magnitude of the induced electromotive force (emf):

1. The number of turns in a coil.

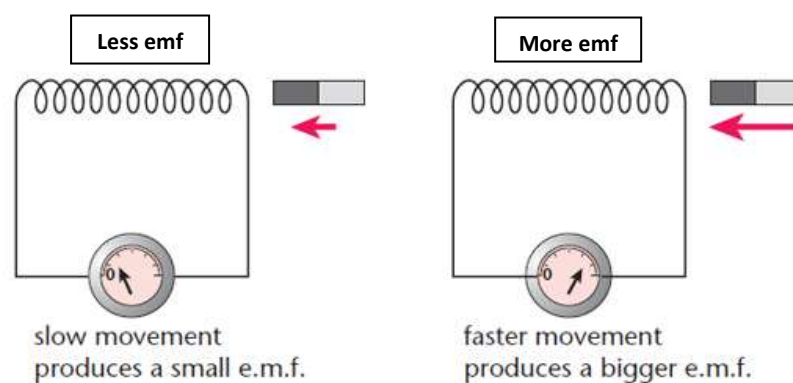
=> As the number of turns or loops increases, the magnitude of the induced emf increases.



Inductors : 4 Steps - Instructables

2. The speed of the relative motion of the coil and magnet

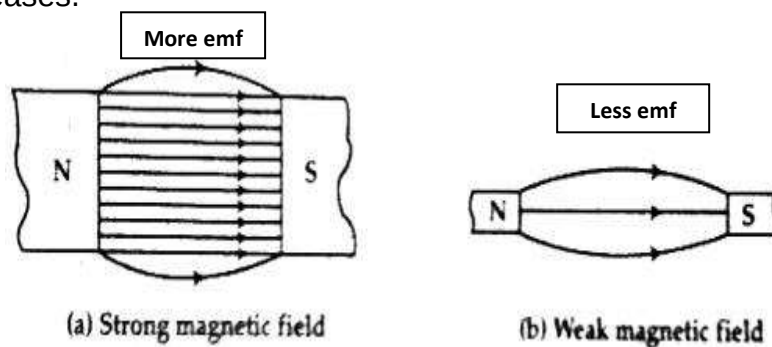
=> As the speed of the relative motion of the coil and magnet increases, the magnitude of induced emf increases.



Electromagnetic induction - A-Level Physics Revision (revisionworld.com)

3. The strength of magnetic field

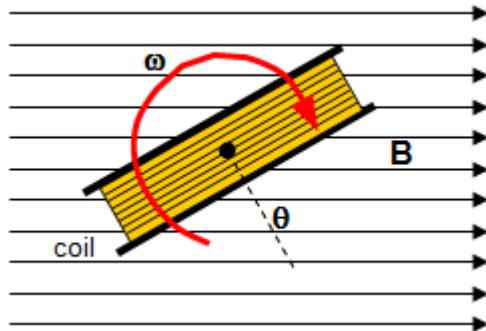
=> As the strength of the magnetic field increases, the magnitude of the induced increases.



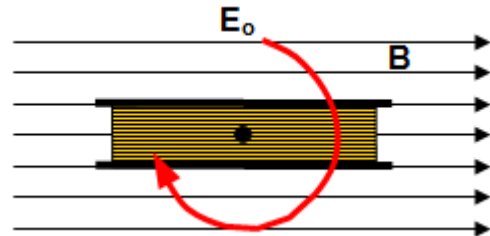
Magnetic Flux - QS Study

4. The rate of change in magnetic flux (the number of cut magnetic lines of force)
 $\Rightarrow$ As the number of cut magnetic lines of force increases, the magnitude of induced emf increases.

Note: The change in magnetic flux is maximum when the angle between the motion of the wire and magnetic field is 90° (perpendicular) and it is zero when the angle between them is 0° (parallel).



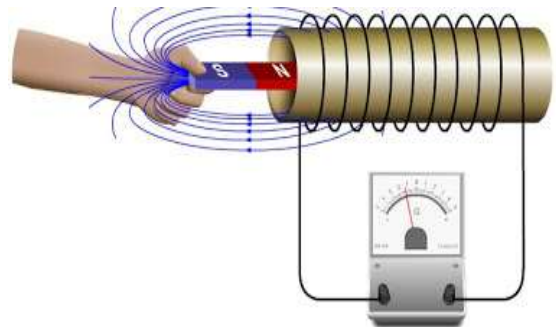
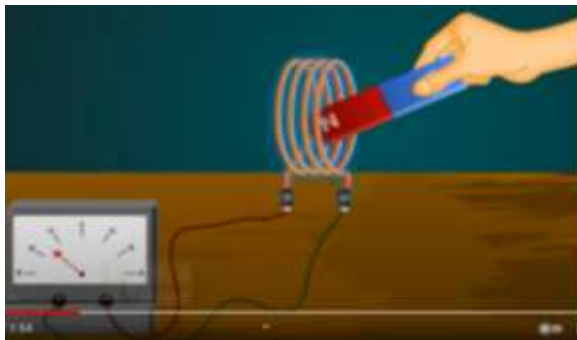
More lines are cut,
greater induced emf



Lesser lines are cut,
lesser induced emf

<https://www.schoolphysics.co.uk>

Likewise, an emf induced by a changing magnetic field can also produce induced current, whether or not there is a path for current to flow. Induced current is directly proportional to induced emf. Similar to induced emf, the magnitude of induced current is also affected by the same factors. Look at the illustrations below, a changing magnetic field produces current in a wire loop as indicated by the deflection of the needle of the galvanometer.

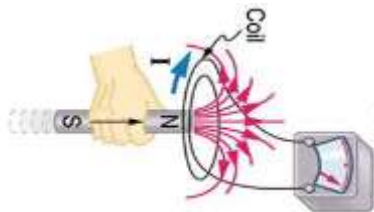
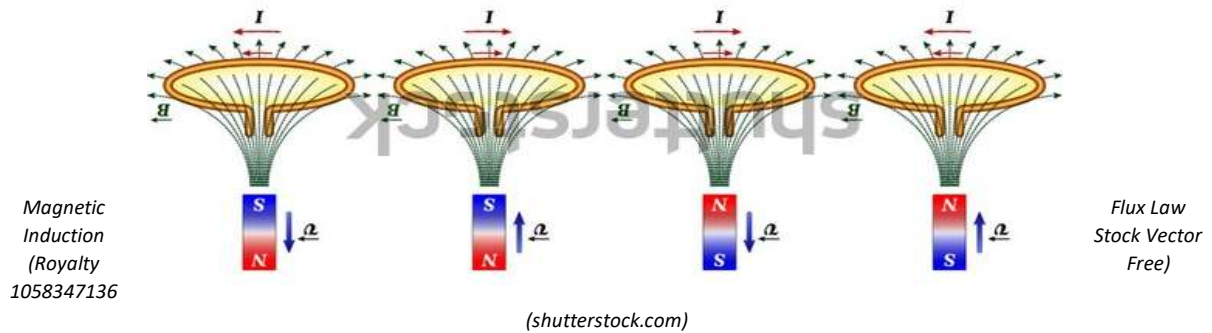


Note: The needle of the galvanometer deflects because of the induced current.

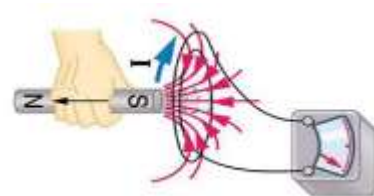


An electric bulb lights up because of induced current.

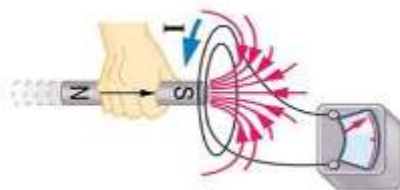
The direction of induced current depends upon the direction of motion of either the magnet or the coil. Look at the direction of the induced current due to the motion of the magnet.



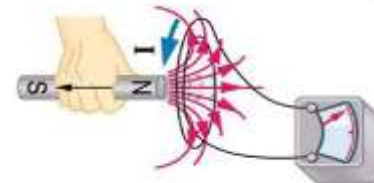
a) When the **north pole** of magnet is moved into the coil, the direction of induced current is **counter clockwise**.



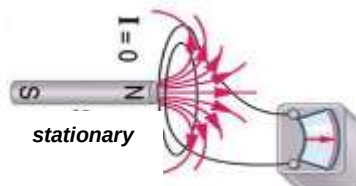
b) When the **north pole** of magnet is moved away from the coil, the direction of induced current is **clockwise**.



c) When the **south pole** of magnet is moved into the coil, the direction of induced current is **clockwise**.



d) When the **south pole** of magnet is moved into the coil, the direction of induced current is **counter clockwise**.

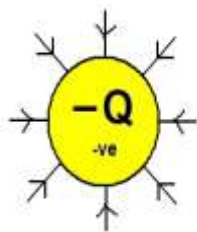
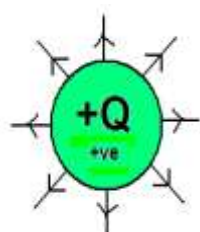


e) When a magnet is held stationary, the induced current is **zero**.

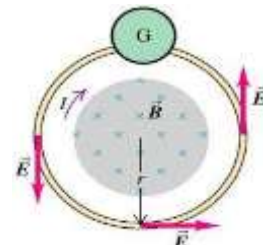
<https://www.texasgateway.org/resource/202-electromagnetic-induction>

Note: The galvanometer reads zero current at the center of the scale and its needle deflects left or right depending on the direction of current.

Similarities and differences between electrostatic electric field and Induced electric field



<https://docbrown.info/ephysics/electricity8.htm>



UY1: Induced Electric Field | Mini Physics - Learn Physics

a) Electrostatic electric field

b) Induced electric field

Electrostatic electric field	Induced electric field
It is caused by stationary charges	It is caused by changing magnetic flux
The field lines do not form closed loops because no electric field line exist inside the charged body.	The field lines form closed loops around the coil of wire.
Its force is conservative because it does no net work over a closed path.	Its force is non-conservative because it does net work in moving a charge over a closed path.
The field $\oint \vec{E} \cdot d\vec{l}$ is equal to zero , therefore, electric potential has meaning.	The field $\oint \vec{E} \cdot d\vec{l}$ is not equal to zero, therefore electric potential has no meaning.
The work or energy difference does not depend on path	The work or energy difference depends on path
The field exerts a force, $F = qE$ on charged particle.	The field also exerts a force $F = qE$ on charged particle.
The field is strong when the field lines are close to each other.	The field is strong when the field lines are close to each other.
Their field lines never cross each other.	Their field lines never cross each other.

Faraday's Law of Induction

Faraday's experiments showed the different factors affecting the induced EMF. According to him, EMF is directly proportional to the **change in magnetic flux** ($\Delta\phi$) and the number of turns (N) of coil. However, it is inversely proportional to the **change in time** (Δt), that is, EMF is greatest when the change in time is smallest. The equation for the EMF induced by a change in magnetic flux is given by the equation,

$$\epsilon = - \frac{N \Delta\phi}{\Delta t}$$

where: ϵ = induced emf (in unit volt, V)
 N = number of turns of coil

$\Delta\phi$ = change in flux

Δt = change in time (in unit seconds, s)

If the magnetic flux linked with a coil of cross-sectional area A , having N turns and at right angle to the magnetic field of flux density B , the total magnetic flux (ϕ) is NBA . Therefore, induced emf can be expressed as,

$$\epsilon = \frac{-N\Delta B}{\Delta t}$$

where: ϵ = induced emf (in unit volt, V)

N = number of turns of coil

A = cross sectional area of coil (in unit m^2)

ΔB = change in magnetic field (in unit tesla, T)

Δt = change in time

Note: If the normal to the plane of loops is directed at an angle θ to a magnetic field B , then the equation of induced emf would be,

$$\epsilon = \frac{-N\Delta B \cos\theta}{\Delta t}$$

where: θ = angle between the magnetic field (B) and the normal (perpendicular) to area (A)

This relationship is known as Faraday's Law of Electromagnetic Induction which states that the emf induced by a change in magnetic flux depends on the change in flux, change in time, and the number of turns of coils. Note that the negative sign only denotes that the induced emf and induced current oppose the original change of magnetic flux.

The electromagnetic induction is a phenomenon in which a changing magnetic flux can induce an electromotive force (emf) in a coil.

Example:

1. A coil of wire of area **2 cm²** containing **20 loops** is situated in a magnetic field. The magnetic field changes from **20 T** to **10 T** in **2 seconds**. What is the emf induced in the coil of wire?

Given: $A = 2 \text{ cm}^2 = 2 \times 10^{-4} \text{ m}^2$

$N = 20$

$\Delta B = 20\text{T} - 10\text{T} = 10\text{T}$

$\Delta t = 2 \text{ seconds}$

Solution:

$$E = N \frac{\Delta\phi}{\Delta t} \Rightarrow \phi = BA$$

$$E = NA \frac{\Delta B}{\Delta t}$$

$$= (20) (2 \times 10^{-4} \text{ m}^2) \frac{(20\text{T} - 10\text{T})}{2 \text{ s}}$$

$$E = 0.02\text{V}$$

2. A coil of wire with **5** turns and a cross-sectional area of **5 cm²** cuts the magnetic field lines at an angle of **30⁰**. It changes the magnetic field strength from **15 T** to **10 T** in **0.5 s**. What is the magnitude of induced emf in the coil?

Given: $A = 5 \text{ cm}^2 = 5 \times 10^{-4} \text{ m}^2$
 $N = 5$
 $\Delta B = 15\text{T} - 10\text{T} = 5\text{T}$
 $\Delta t = 0.5 \text{ seconds}$
 $\Theta = 30^0$

Solution:

$$E = N \frac{\Delta \phi \cos \Theta}{\Delta t} \quad \Rightarrow \quad \phi = BA$$

$$E = \frac{NA \Delta B \cos \Theta}{\Delta t}$$

$$= \frac{(5) (5 \times 10^{-4} \text{ m}^2) (15\text{T} - 10\text{T}) (\cos 30^0)}{0.5 \text{ s}}$$

$$= \frac{(25 \times 10^{-4} \text{ m}^2) (5 \text{ T}) (0.87)}{0.5 \text{ s}}$$

$$E = 2.175 \times 10^{-2} \text{ V}$$

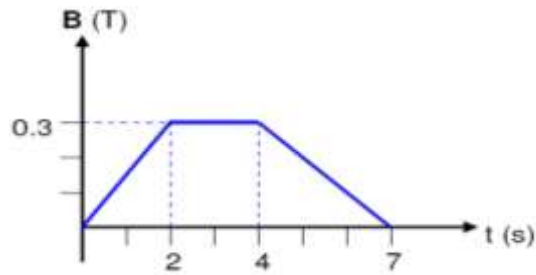
APPLICATION

Direction: Solve the following problems and show all your solutions.

1. A **50-turn** coil has a diameter of **15 cm**. The coil is placed in a magnetic field of magnitude **0.50 T** so that the face of the coil and the magnetic field are perpendicular. Find the magnitude of the emf induced in the coil if the magnetic field is reduced to zero in **10 s**. (Hint: The area of a circular coil is **$A = \pi r^2$**)

2. A magnetic field passes through a stationary wire loop and its magnitude changes in time based on the graph below. The loop consists of **20 turns** of wire and has an

area of 0.005 m^2 . Determine the induced emf in time interval $t = 0$ second to $t = 2$ seconds.



Study.com/academy/answer

That was excellent! Keep it up!

POST-TEST

Directions: Read each item carefully and identify what is asked or described. Encircle the letter of the correct answer.

1. Which of the following is NOT true about induced electric field?
 - A. It is caused by the changing magnetic field.
 - B. Its work or energy difference depends on path.
 - C. Its field $\oint \mathbf{E} \cdot d\mathbf{l}$ is not equal to zero, therefore electric potential has no meaning.
 - D. Its force is conservative because it does net work in moving a charge over a closed path.

2. The following are factors involved in the induced emf of a coil of wire EXCEPT
 - A. The induced emf is directly proportional to N , the total number of turns in the coil.
 - B. The induced emf is directly proportional to A , the area of cross-section of the coil.
 - C. The induced emf is directly proportional to instant t , the time of changing magnetic flux.
 - D. The induced emf is directly proportional to B , the strength of the magnetic field in which the coil is rotating.

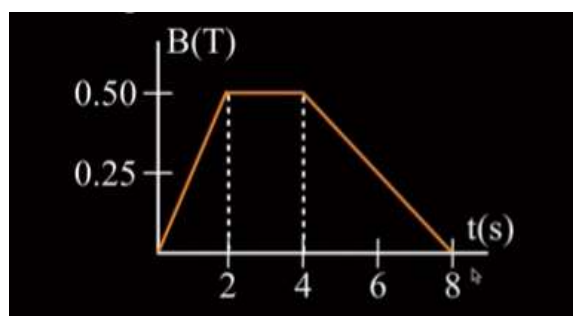
3. When the north pole of a magnet is moved into a coil, the direction of the induced current in the coil as seen from the magnet side is

A. Clockwise	C. Clockwise then counter clockwise
B. Counter clockwise	D. Counter clockwise then clockwise

4. A loop lies in a magnetic field, with the field perpendicular to the plane of the loop. Which does NOT induce a current in the loop?
 - A. Increasing the size of the loop.
 - B. Changing the direction of the field.
 - C. Rotating the loop around the magnetic field.
 - D. Sliding the loop perpendicular to the field vectors.
5. Faraday's law of induction states that
 - A. the direction of the induced emf is such that it opposes the change of flux.
 - B. the magnitude of the induced emf is directly proportional to the rate of change of flux.
 - C. the magnitude of the induced emf is inversely proportional to the rate of change of flux.
 - D. the magnitude of the induced emf is directly proportional to the square of the rate of change of flux.
6. In the demonstration where a magnetic north pole is brought near a loop, which is TRUE about the induced current if we use the same speed but switch to the south end of the magnet?
 - A. The magnitude and direction are the same.
 - B. The magnitude is the same but the direction is reversed.
 - C. The magnitude is now smaller and in the opposite direction.
 - D. The magnitude is now larger and in the same direction as before.
7. Doubling the number of loops of wire in a coil produces what kind of change on the induced emf, assuming all other factors remain constant?
 - A. The induced emf is twice as much.
 - B. The induced emf is 3 times as much.
 - C. The induced emf is 4 times as much.
 - D. The induced emf remains the same.
8. A wire loop is being pulled through a uniform magnetic field. Which of the following statement is TRUE?
 - A. The induced current is constant.
 - B. There is no induced current produced.
 - C. The direction of induced current is clockwise.
 - D. The direction of induced current is counter clockwise.
9. Which of the following is NOT true about induced emf?
 - A. It is caused by a constant magnetic field.
 - B. It is directly proportional to induced current.
 - C. It is always opposite to the direction of magnetic flux.
 - D. It is only produced by the motion perpendicular to the field.

10. What does emf stand for?
- A. Electromotive force
B. Electro mobile force
C. electromagnetic factor
D. electromagnetic force
11. The coil of wire containing **two loops** with a radius of **6 cm**. is in a **1.8 T** magnetic field. The plane of the coil of wire forms **60°** with the magnetic field. The loop is removed from the magnetic field over a time of **0.25 s**. What is the induced emf in the loop? (Hint: The area of the circular coil is **$A = \pi r^2$**)
- A. $4 \times 10^{-2} \text{ V}$
B. $6 \times 10^{-2} \text{ V}$
C. $7 \times 10^{-2} \text{ V}$
D. $8 \times 10^{-2} \text{ V}$
12. The following are the similarities of electrostatic electric field and induced electric field EXCEPT
- A. their field lines form closed loops.
B. their field lines never cross each other.
C. their field exerts a force, $F = qE$ on charged particle.
D. their field is strong when the field lines are close together.

For questions #13-15: The magnetic field through a single loop of wire, **10 cm** in radius and with a resistance of 7.5 ohms changes with time as shown in the graph below.



13. What is the induced emf in the loop of wire from $t = 4$ seconds to $t = 6$ seconds?
- A. 0 V
B. $2 \times 10^{-3} \text{ V}$
C. $4 \times 10^{-3} \text{ V}$
D. $5 \times 10^{-3} \text{ V}$
14. What is the induced emf in the loop of wire from $t = 0$ second to $t = 2$ seconds?
- A. $7 \times 10^{-3} \text{ V}$
B. $8 \times 10^{-3} \text{ V}$
C.
D. $8 \times 10^{-2} \text{ V}$
15. What is the induced emf in the loop of wire from $t = 2$ seconds to $t = 4$ seconds?
- A. 0 V
B. 0.25 V
C. 0.5 V
D. 2 V

GLOSSARY

The following terms used in this module are defined as follows for you to be guided:

INDUCED EMF	short-lived voltage generated by a conductor or coil moving in a magnetic field.
MAGNETIC FLUX	measurement of the amount of magnetic field lines through a given area.
INDUCTION	the generation of an electric current by a varying magnetic field.
ELECTROMOTIVE FORCE	the voltage generated by a battery or by changing magnetic field. It is not actually a force.

ANSWER KEY

PRE-TEST

- | | | |
|------|-------|-------|
| 1. A | 6. B | 11. A |
| 2. A | 7. D | 12. D |
| 3. B | 8. B | 13. B |
| 4. A | 9. C | 14. A |
| 5. B | 10. D | 15. C |

REVIEW OF THE PREVIOUS LESSON

- | | |
|----------------------|--|
| 1. A) 22 J
B) 7 V | 2. A) Charge B
B) Charge B
C) Charge B |
|----------------------|--|

PRESENTATION OF THE LESSON

Activity

- A. Answers may vary (The line graph is straight)
 B. 1) Greater also
 2) Smaller also
 C.

Time interval (in unit s)	Change in magnetic field $(\Delta B = B_{\text{final}} - B_{\text{initial}})$ (in unit Tesla, T)	Rate of change in magnetic flux $\Delta \phi = A \Delta B$ (in unit weber, Wb)	Induced Emf $\text{emf} = \frac{N \Delta \phi}{\Delta t}$ (in unit volt, V)
0 s – 3.0 s	0.4 T	$3.14 \times 10^{-3} \text{ Wb}$	$1.05 \times 10^{-3} \text{ V}$
3.0 s – 6.0 s	0 T	0 Wb	0 V
6.0 s – 9.0 s	0.2 T	$1.57 \times 10^{-3} \text{ Wb}$	$5.2 \times 10^{-4} \text{ V}$

Analysis

- | | |
|---|--|
| A. 1) Straight line graph
2) Number of turns is proportional to induced emf.
3) Increases | 4) Decreases
5) Directly proportional |
| B. 1) Greater 2) Smaller 3) Directly proportional | |
| C. 1) Increases
2) Increases, Decreases
3) Zero | 4) Zero
5) Directly proportional
6) The greater the time, the lesser the induced emf |

APPLICATION

- 1) $\epsilon = 4.42 \times 10^{-2} \text{ V}$
- 2) $\epsilon = 1.5 \times 10^{-2} \text{ V}$

POST-TEST

- | | | |
|------|-------|-------|
| 1. D | 6. B | 11. D |
| 2. C | 7. A | 12. A |
| 3. B | 8. B | 13. C |
| 4. D | 9. A | 14. B |
| 5. B | 10. A | 15. A |

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